



DG INTERNS HUB
CYBERSECURITY INTERNSHIP

WEEK 2 ASSIGNMENT REPORT

Advanced Networking, Analysis & Security Concepts

NETWORK ANALYSIS	SECURITY FUNDAMENTALS	SAFE LAB PRACTICE
Wireshark & DNS/HTTPS	Ports, devices & attacks	Nmap & Packet Tracer

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Report

Week 2 Assignment

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Objective & Scope

This report documents the Week 2 cybersecurity internship work covering advanced networking concepts, real packet analysis, authorized network scanning, LAN configuration, and a safe conceptual MITM simulation.

Safety & Ethics

Only free tools, the assignment-authorized Nmap target, and local Packet Tracer simulations were used. No real MITM interception or ARP-spoofing attack was performed.

Assignment Structure

Task	Topic	Category
01	Ports & Services	Theory
02	Networking Devices	Theory
03	Network Security Basics	Theory
04	Packet Analysis - Wireshark	Practical
05	Network Scanning - Nmap	Practical
06	LAN Setup - Packet Tracer	Practical
07	MITM Concept - Safe Simulation	Practical

Ports & Services

A network port is a logical communication endpoint used to direct traffic to a particular service or application. TCP is connection-oriented and focuses on reliable, ordered delivery; UDP is connectionless and uses lower overhead.

Port	Protocol	Service	Primary Usage
20	TCP	FTP Data	FTP data transfer
21	TCP	FTP Control	FTP commands and session setup
22	TCP	SSH / SFTP	Secure remote access and file transfer
23	TCP	Telnet	Unencrypted remote terminal access
25	TCP	SMTP	Sending email
53	TCP/UDP	DNS	Domain name resolution
67 / 68	UDP	DHCP	Dynamic IP address configuration
69	UDP	TFTP	Lightweight file transfer
80	TCP	HTTP	Unencrypted web traffic
110	TCP	POP3	Email retrieval
123	UDP	NTP	Time synchronization
139	TCP	NetBIOS	Legacy Windows sharing
143	TCP	IMAP	Server-side email access
161 / 162	UDP	SNMP	Monitoring and alerts
389	TCP/UDP	LDAP	Directory service queries
443	TCP	HTTPS	Encrypted web traffic
445	TCP	SMB	Windows file/printer sharing
636	TCP	LDAPS	Secure LDAP
993 / 995	TCP	IMAPS / POP3S	Secure email retrieval
1433 / 3306 / 3389	TCP	MSSQL / MySQL / RDP	Database and remote desktop services

Key takeaway

Knowing common ports helps with traffic analysis, firewall policy review, network monitoring, and interpreting scan results.

Networking Devices

Switches, routers, and firewalls have different responsibilities. A switch primarily connects devices inside a LAN, a router connects different IP networks, and a firewall enforces security policy on network traffic.

Device	Working	Real-world Example
Switch	Learns MAC addresses and forwards Ethernet frames to the appropriate port inside a LAN.	Office PCs and printers connected to one local network.
Router	Uses IP addresses and routing information to forward packets between networks.	A home router connecting the LAN to the Internet.
Firewall	Inspects traffic against configured security rules and permits or blocks connections.	An organization blocking unauthorized inbound services.

Router vs. Switch

Feature	Switch	Router
Primary role	Connects devices in the same LAN	Connects different networks
Typical OSI layer	Layer 2 - Data Link	Layer 3 - Network
Addressing	MAC address	IP address
Data unit	Frame	Packet
Key takeaway Switching handles local frame delivery; routing moves packets between different networks.		

Network Security Basics

Network security protects infrastructure, connected devices, and data from unauthorized access, misuse, disruption, and modification. A core security model is the CIA Triad: Confidentiality, Integrity, and Availability.

Threat	Concept	Basic Prevention
Man-in-the-Middle (MITM)	An attacker attempts to position themselves between communicating parties to observe or manipulate traffic.	HTTPS/TLS, trusted VPNs, certificate checks, secure networks
Denial of Service (DoS)	A target is overwhelmed with traffic or requests so legitimate users cannot access resources.	Rate limiting, monitoring, WAF/IPS, redundancy
Packet Sniffing	Network packets are captured and inspected; unencrypted traffic may expose sensitive information.	Encryption, secure protocols, segmentation
Security principle	Encryption protects data in transit, while monitoring and access controls help reduce exposure and detect abnormal activity.	

Packet Analysis with Wireshark

Tool: Wireshark Interface: Wi-Fi

Live network traffic was captured and filtered to identify DNS activity and HTTPS/TLS communication. The screenshots below document the practical workflow and observed packet information.

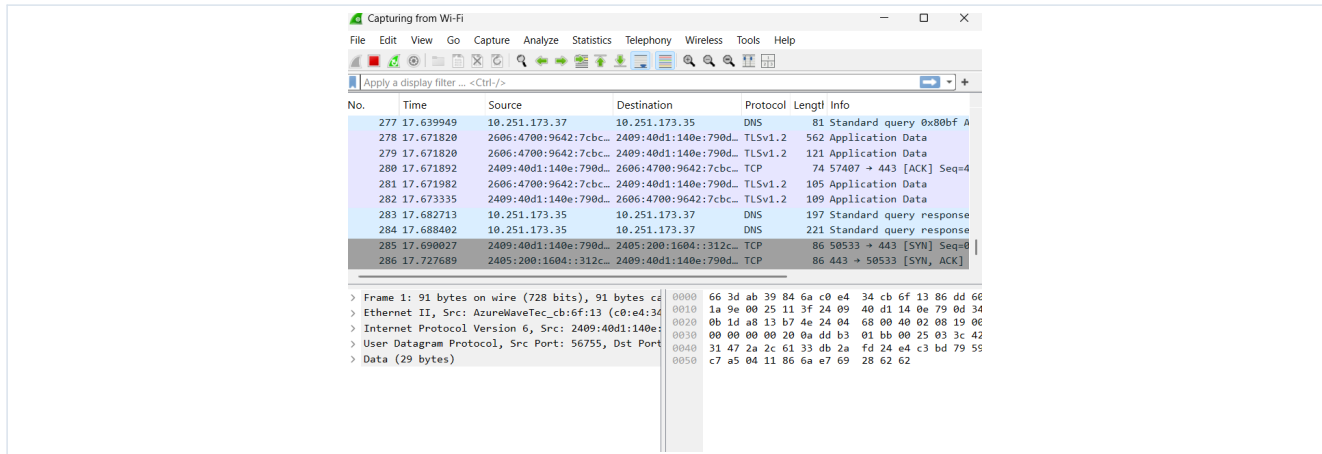


Figure 1. Live packet capture on the Wi-Fi interface.

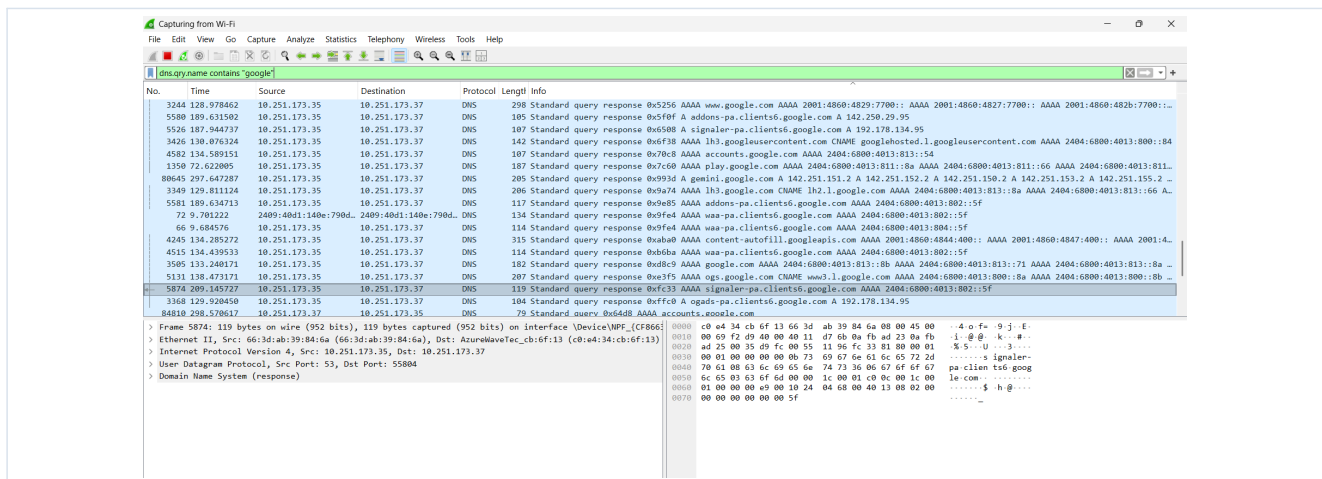


Figure 2. DNS filter showing Google-related DNS traffic.

HTTPS / TLS Traffic

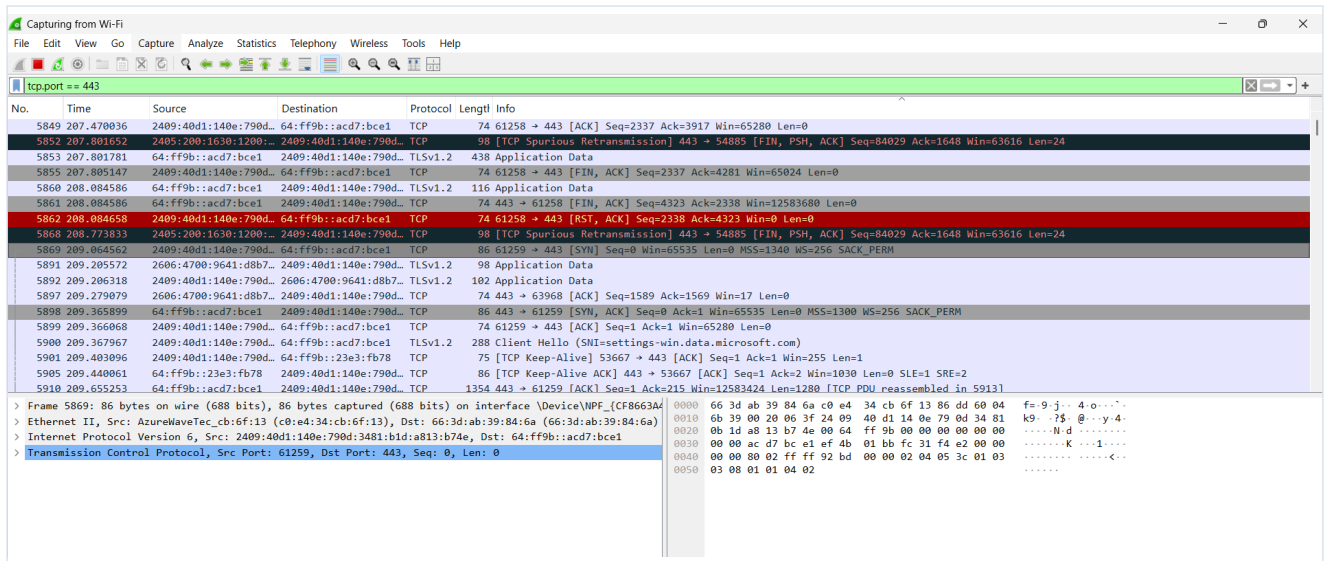


Figure 3. TCP port 443 filter showing HTTPS/TLS traffic and source/destination addressing.

Observation

The capture showed TCP connections using port 443, TLSv1.2 application data, and packet-level source/destination addressing. DNS filters also exposed query/response activity for Google-related domains.

Network Scanning with Nmap

Authorized target: scanme.nmap.org

Command 1: nmap scanme.nmap.org

```
C:\Users\Akash>nmap scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-10 13:33 +0530
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.29s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open       ssh
25/tcp    filtered   smtp
80/tcp    open       http
135/tcp    filtered   msrpc
139/tcp    filtered   netbios-ssn
445/tcp    filtered   microsoft-ds
9929/tcp   open       nping-echo
31337/tcp  open       Elite

Nmap done: 1 IP address (1 host up) scanned in 13.16 seconds
```

Figure 4. Basic Nmap scan of the assignment-authorized target.

Basic scan result

The host was up. Open ports included 22/tcp (SSH), 80/tcp (HTTP), 9929/tcp (nping-echo), and 31337/tcp. Ports 25, 135, 139, and 445 appeared filtered.

Service & Version Detection

Command 2: `nmap -sV scanme.nmap.org`

```
C:\Users\Akash>nmap -sV scanme.nmap.org
Starting Nmap 7.991 ( https://nmap.org ) at 2026-09-10 13:35 +0530
Warning: Hostname scanme.nmap.org resolves to 2 IPs. Using 45.33.32.156.
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.35s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
445/tcp   filtered  microsoft-ds
9929/tcp  open      nping-echo   Nping echo
31337/tcp open      tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.95 seconds
```

Figure 5. Nmap service/version detection output.

Service detection

The scan reported OpenSSH 6.6.1p1 on port 22, Apache httpd 2.4.7 on port 80, Nping echo on port 9929, tcpwrapped on port 31337, and Linux service information.

Cisco Packet Tracer - LAN Setup

Topology: 2 PCs + 1 Cisco 2960 switch

Addressing: PC1 = 192.168.1.10/24 PC2 = 192.168.1.11/24

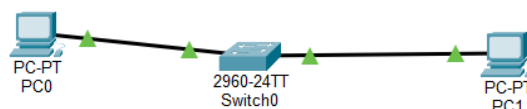


Figure 6. Final two-PC LAN topology with active switch links.

Connectivity was verified by sending ICMP echo requests to 192.168.1.11 from the Packet Tracer command prompt.

```
PC1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\> ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=5ms TTL=128
Reply from 192.168.1.11: bytes=32 time=13ms TTL=128
Reply from 192.168.1.11: bytes=32 time=14ms TTL=128
Reply from 192.168.1.11: bytes=32 time=9ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 14ms, Average = 10ms

C:\>
```

Figure 7. Successful ping to 192.168.1.11: 4 sent, 4 received, 0% packet loss.

Result

The successful ping confirms communication between the two PCs in the 192.168.1.0/24 LAN.

MITM Concept - Safe Simulation

Planned addressing: Victim = 192.168.1.10 Attacker = 192.168.1.20 Router/Gateway = 192.168.1.1

This lab illustrates the logical position of a potential Man-in-the-Middle attacker. It is a concept-only simulation; no real interception or ARP-spoofing attack was executed.

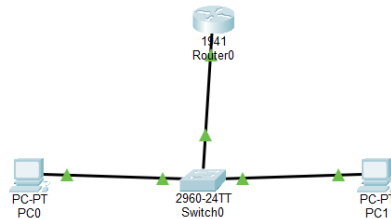


Figure 8. Safe MITM concept topology using two PCs, a switch, and a router.

```
C:\> ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=10ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time=12ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 5ms

C:\>
```

Figure 9. Packet Tracer connectivity test showing four replies and 0% packet loss.

Area	Summary
How MITM works	An attacker attempts to place themselves logically between a victim and another endpoint or gateway.
Real-world example	An insecure public Wi-Fi network.
Risks	Data theft, credential exposure, session interception, modification of unprotected traffic.
Prevention	HTTPS/TLS, trusted VPNs when appropriate, secure networks, and certificate verification.

Week 2 Learning Outcomes

This assignment combined theory with practical network analysis. I reviewed common ports and services, compared core networking devices, studied basic attack concepts, analyzed real packets in Wireshark, performed authorized Nmap scanning, built a LAN in Packet Tracer, and demonstrated the MITM concept in a safe simulation.

Wireshark DNS & HTTPS/TLS analysis	Nmap Ports & service detection	Packet Tracer LAN & MITM simulation
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Submission note

All practical screenshots included in this report were captured during the Week 2 practical work. The Nmap target used was the safe target specified in the assignment.

END OF REPORT